Modeling Bimanual Perceptual Interactions in Macaque Monkeys

Bayesian Hierarchical Modeling

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Outline

Task and Data
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- Model Structure
- Synthetic Data
- Real Data

Model B: Sessions Hierarchical
- Model Structure
- Synthetic Data
- Real Data

Model C: Distractor Amplitude as a Covariate
- Model Structure
- Synthetic Data
- Real Data

Model Comparison and Recovery
- Predictive Performance
- Model Recovery: B vs A
- Model C vs Model B

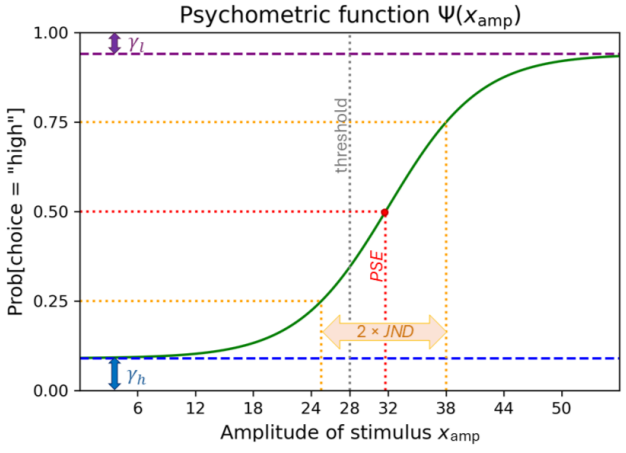
Summary

The task: 2AFC categorization of vibration amplitude

- stimulus $x_{\text{stim}} \in \{6, 12, 18, 24, 32, 38, 44, 50\}\ \mu m$
  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\ \mu m$
  - $x_{\text{d}} \in [6, 32]\ \mu m$, value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 μm
  - "low" response: correct for $x_{\text{stim}} < 28\ \mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\ \mu m$
- correct responses are rewarded with water or juice

Psychometric function: response $r \sim \text{Bernoulli}(p = \psi)$

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}})$$

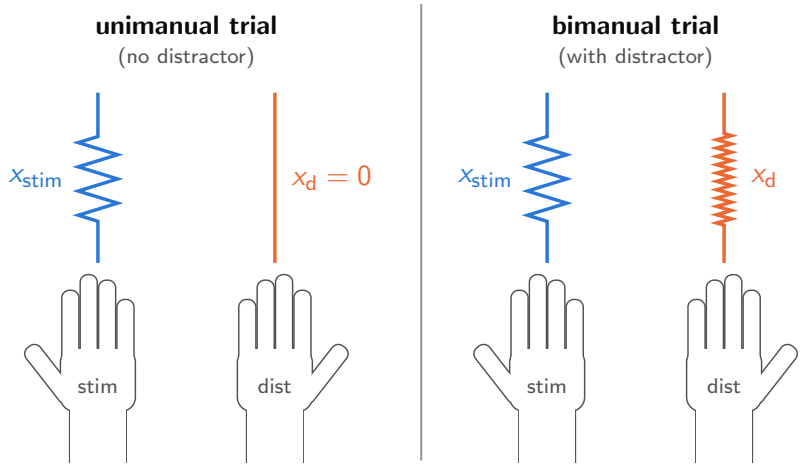


Parameters

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}}), \qquad \vec{\theta} = (\beta_0, \beta_1, \gamma_h, \gamma_l)$$

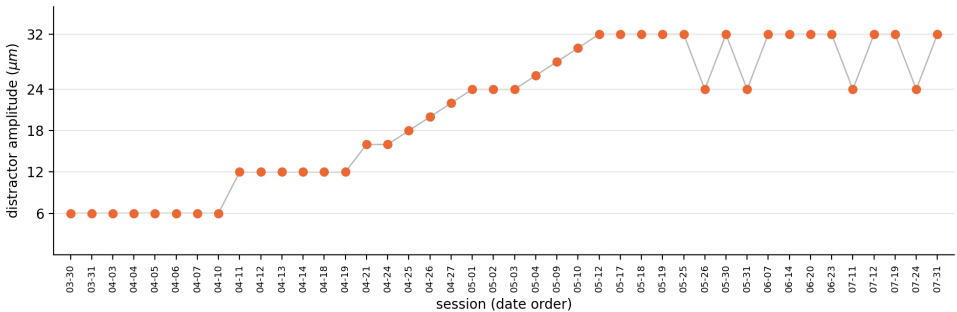
- γ_h: lapse rate for "high"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"high"})$, even to the weakest stimuli
  - lower (left) asymptote of ψ
- γ_l: lapse rate for "low"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"low"})$, even to the strongest stimuli
  - $1 - \gamma_l$ = upper (right) asymptote of ψ
- logit^{-1}: the logistic sigmoid, $\text{logit}^{-1}(z) = \dfrac{1}{1 + \exp(-z)}$
- PSE: stimulus where responses of "high" and "low" are equally likely
  - x_{stim} such that $\psi(x_{\text{stim}}) = 0.5$
  - represents bias
- JND: how much the stimulus needs to change to be noticed half the time
  - half the distance between $\psi(x_{\text{stim}}) = 0.25$ and 0.75
  - represents sensitivity

The task: one trial



The data

- One monkey (Sirius), 12,937 trials, over 42 sessions of about 308 trials
- 4 groups: hand $\times$ (unimanual, bimanual)
- Each session uses a single distractor amplitude on its bimanual trials:

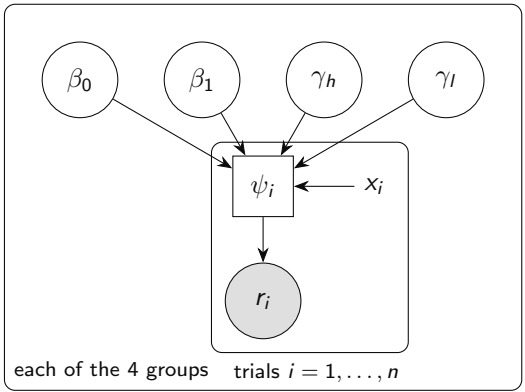


Three models

Model	Sessions	Distractor
A	pooled: one curve per group	uni and bi fit separately
B	hierarchical: one curve per group and session	uni and bi fit separately
C	hierarchical: one curve per hand and session	covariate: shift $\beta_2 x_{\text{dist}}$

For each model: convergence study on synthetic data, then the fit to the real data.

Model A: graphical model



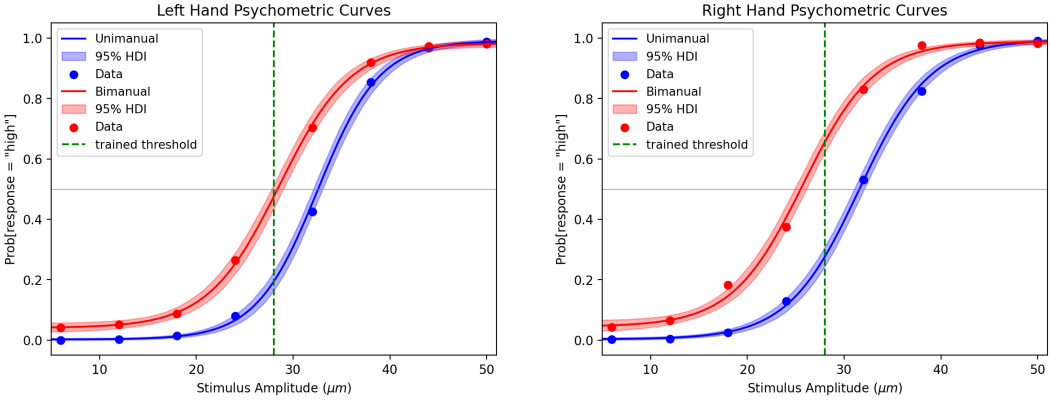
- circles: random variables
- shaded: observed
- rectangles: deterministic
- x_i: fixed covariate
- outer plate: the same model, fit independently to each group

Model A on synthetic data: what is recoverable

At the size of the real data (about 3,230 trials per group):

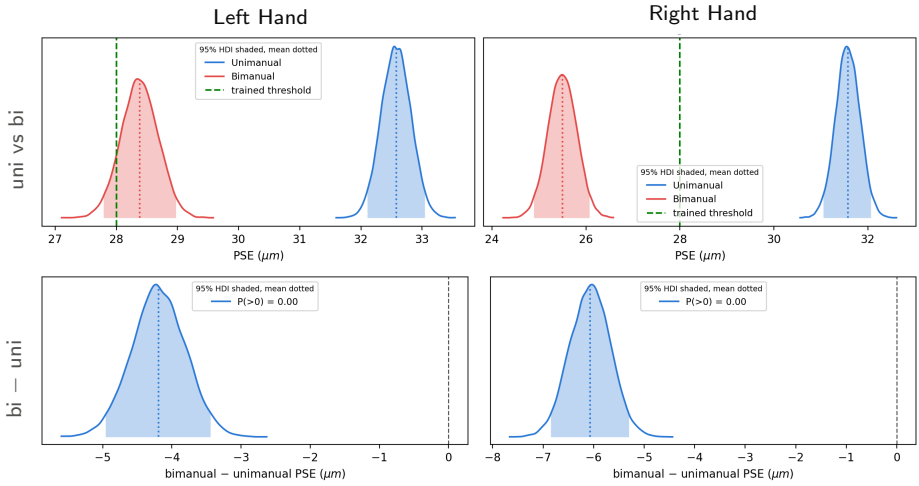
- **Recoverable:** PSE, JND, γ_h, unbiased and with calibrated 95% HDIs
- **Not reliably recoverable:** small γ_l, biased upward
- **Caveat:** data simulated with one curve per group, so this says nothing about variation between sessions

Model A on the real data: fitted curves



Unimanual in blue, bimanual in red, with the observed proportions pooled over sessions; dashed: the trained threshold of 28 μm.

Model A on the real data: effect of the distractor on PSE



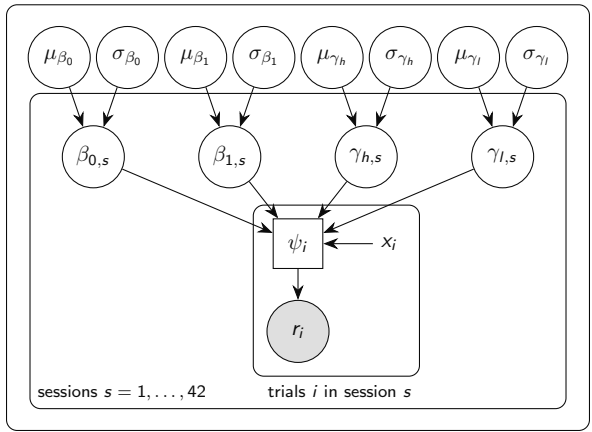
Unimanual Bimanual 95% HDI shaded, mean dotted; green dashed: trained threshold

Model A: conclusions

- The pooled data are reproduced at every amplitude
- Effect of the distractor (bimanual − unimanual):
  - **PSE:** lowered, by about 4 μm (left) and 6 μm (right)
  - **JND:** raised for the left hand; unclear for the right
  - γ_h**:** raised, by about 0.04
  - γ_l**:** no clear effect

$\Rightarrow$ A curve for every session: Model B

Model B: graphical model



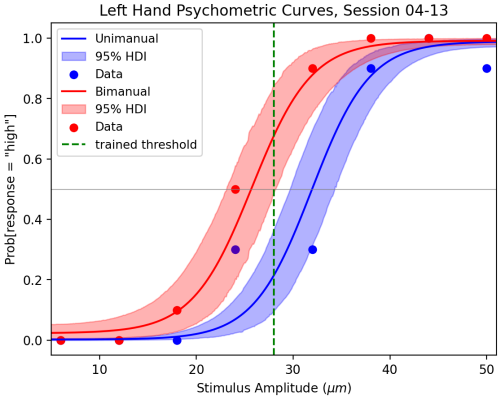
Circles: random variables (shaded: observed); rectangle: deterministic; x_i: fixed covariate. The same model is fit independently to each group.

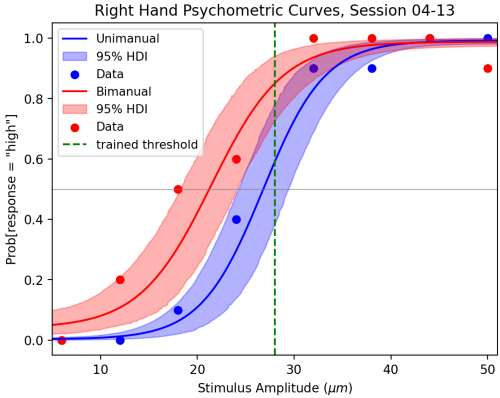
Model B on synthetic data: what is recoverable

At the size of the real data (42 sessions of about 308 trials), at values like the real ones:

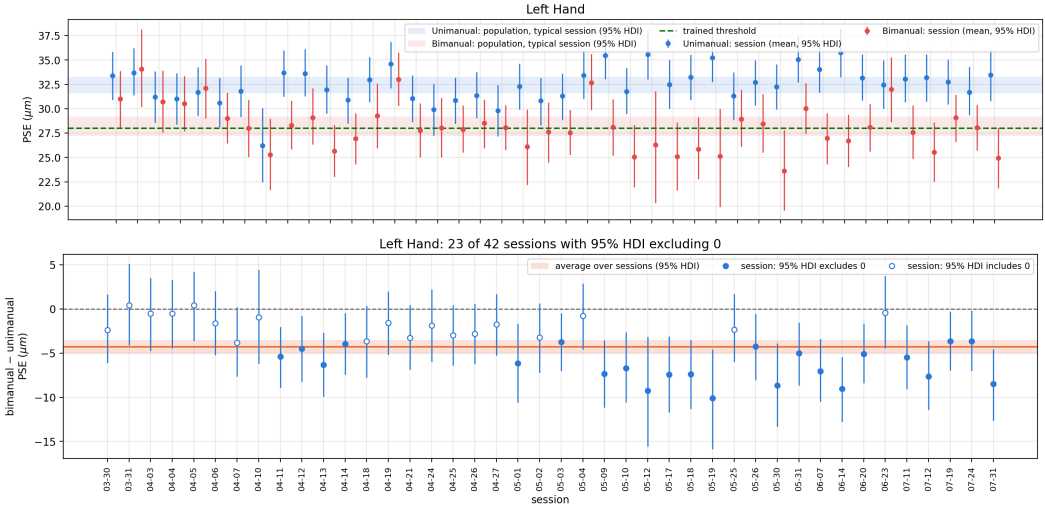
- **Recoverable:** each session's PSE and JND, and how much they vary between sessions
- **Weakly determined:** the lapse rates, especially session by session
- **Not identified:** how much the lapse rates vary between sessions; the estimate reflects the prior

Model B on the real data: fitted curves in one session (04-13)

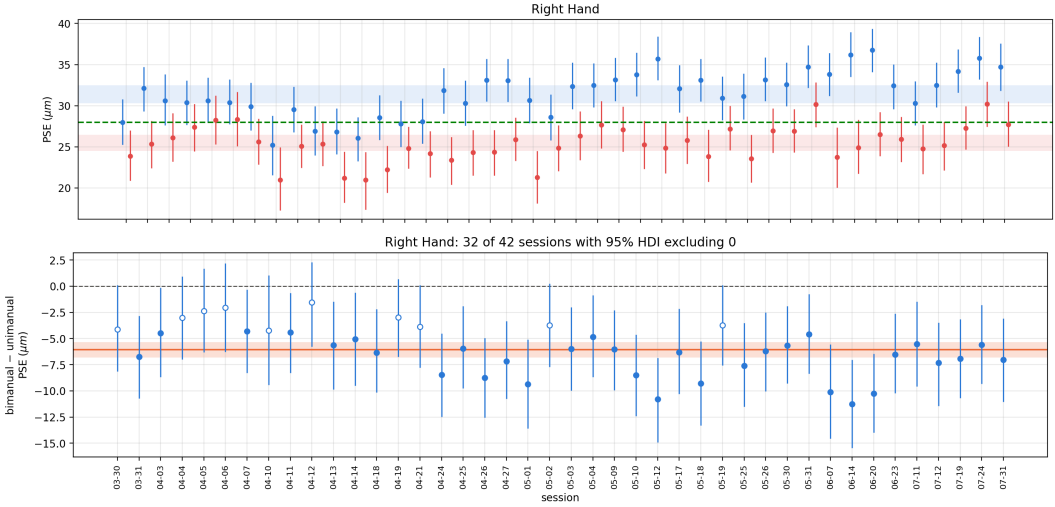




Model B on the real data: PSE, left hand



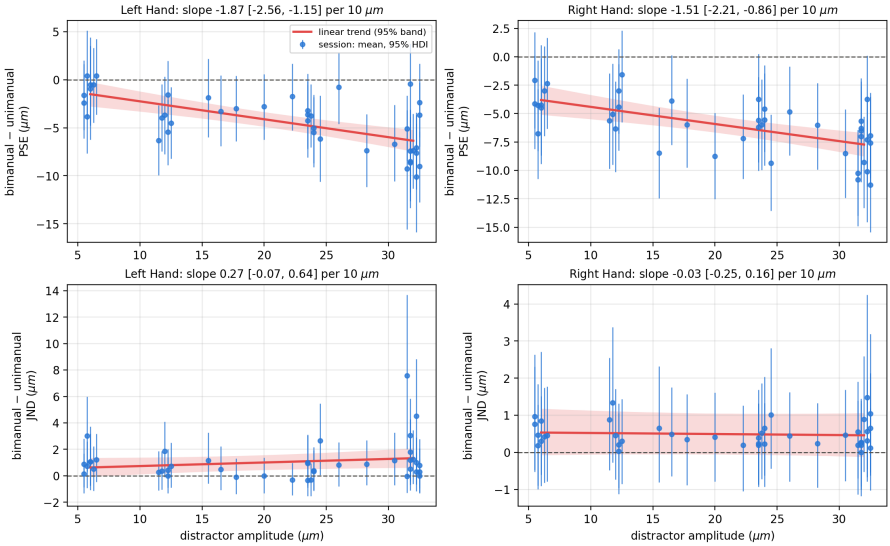
Model B on the real data: PSE, right hand



● unimanual, ● bimanual: session mean and 95% HDI; bands: typical session; green dashed: trained threshold

bi − uni: filled if the session's 95% HDI excludes 0; orange band: average over sessions

Model B: PSE shift vs distractor amplitude (exploratory)



Model B: conclusions

- Reliable fit; no systematic misfit, pooled or within sessions
- Effect of the distractor (bimanual − unimanual), average over sessions:
  - **PSE:** lowered, by about 4 μm (left) and 6 μm (right), in nearly every session
  - **JND:** raised for the left hand, driven by a few sessions (possibly lapses absorbed into the slope)
  - γ_h**:** raised; γ_l**:** no effect
- The PSE shift grows with distractor amplitude (exploratory; amplitude is confounded with session order)

$\Rightarrow$ Put the distractor amplitude into the model: Model C

Model C: including the distractor as a covariate

Psychometric function:

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}\left(\beta_0 + \beta_1 x_{\text{stim}}\right)$$

Suppose the distractor has an impact on the perceived stimulus as follows:

$$x_{\text{perc}} = x_{\text{stim}} + c\, x_{\text{dist}}$$

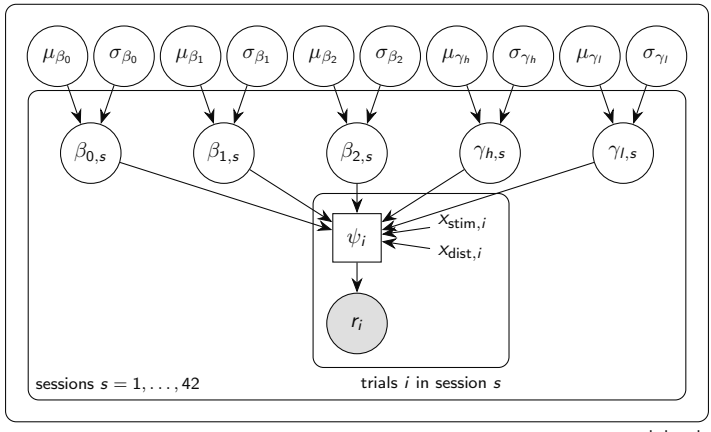
Then:

$$\begin{aligned}\beta_0 + \beta_1 x_{\text{perc}} &= \beta_0 + \beta_1(x_{\text{stim}} + c\, x_{\text{dist}}) \\ &= \beta_0 + \beta_1 x_{\text{stim}} + \beta_1 c\, x_{\text{dist}} \\ &= \beta_0 + \beta_1 x_{\text{stim}} + \beta_2\, x_{\text{dist}}\end{aligned}$$

Considering x_{dist} as a covariate:

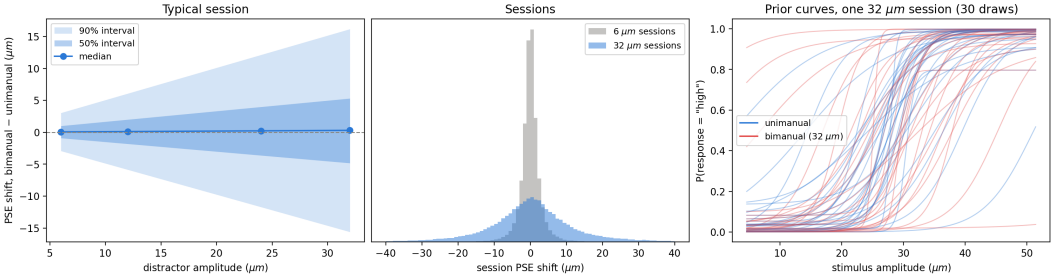
$$\psi(x_{\text{stim}}, x_{\text{dist}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}\left(\beta_0 + \beta_1 x_{\text{stim}} + \beta_2\, x_{\text{dist}}\right)$$

Model C: graphical model



Circles: random variables (shaded: observed); rectangle: deterministic; $x_{\text{stim},i}$, $x_{\text{dist},i}$: fixed covariates. The same model is fit independently to each hand, with unimanual and bimanual trials together.

Model C: prior predictive check of the distractor effect



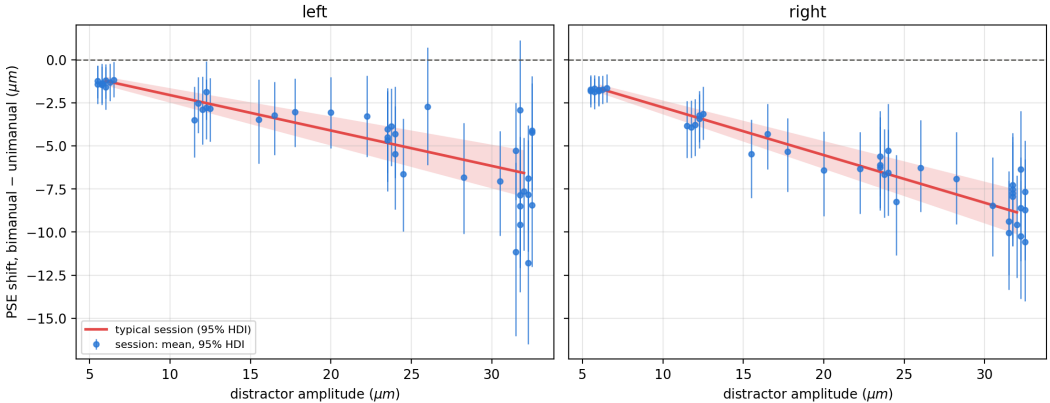
Model C drawback: the PSE shift $-\beta_2 x_{\text{dist}}/\beta_1$ is tied to the distractor amplitude, so weak distractors can only shift the PSE slightly

Model C on synthetic data: what is recoverable

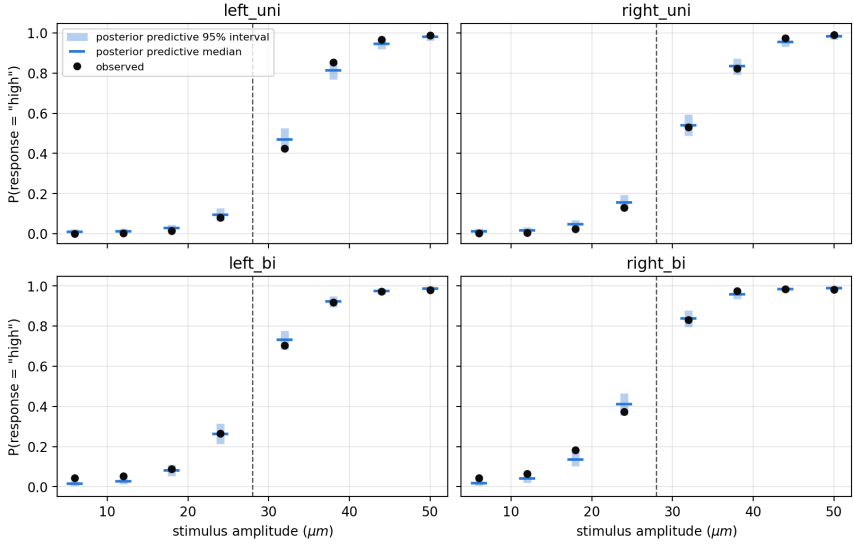
At the size of the real data, if the data follow Model C:

- **Recoverable:** each session's PSE in both conditions, and the distractor effect β_2 of a typical session
- **Weakly determined:** β_2 in sessions with a weak distractor, which is pulled toward the typical value
- **Not tested:** data that break Model C's assumptions (shared lapse rates and slope, a shift proportional to amplitude)

Model C: the PSE shift grows with distractor amplitude



Model C on the real data: misfit at the weakest stimuli



Model C: conclusions

- **PSE:** lowered by the distractor, more the stronger it is: a typical session shifts 1–2 μm at a 6 μm distractor, 6.5–9 μm at 32 μm
- **JND, γ_h, γ_l:** shared by both conditions, so the model cannot show an effect on them
- The misfit sits where its assumptions meet the data:
  - shared lapse rates: too few "high" responses predicted to weak stimuli on bimanual trials
  - a shift proportional to amplitude: on the right hand, even a 6 μm distractor shifts responses more than Model C allows

$\Rightarrow$ Which model describes the data best? Model comparison

Leave-one-out cross-validation

How well each model predicts each trial when that trial is left out of the fit (higher is better):

Model	difference from B (SE)
A: sessions pooled	−164.1 (19.2)
B: sessions hierarchical	**best**
C: distractor as covariate	−37.1 (11.9)

- B beats A clearly, despite many more parameters
- B beats C, although C needs fewer parameters

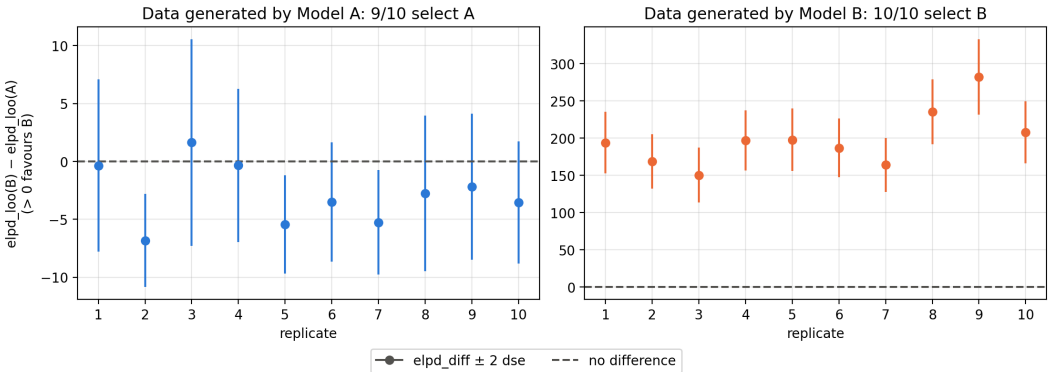
Model recovery: LOO picks the generating model

Is B's advantage real, or just flexibility? Simulate datasets from each model on the real design, fit both, and see which one LOO picks:

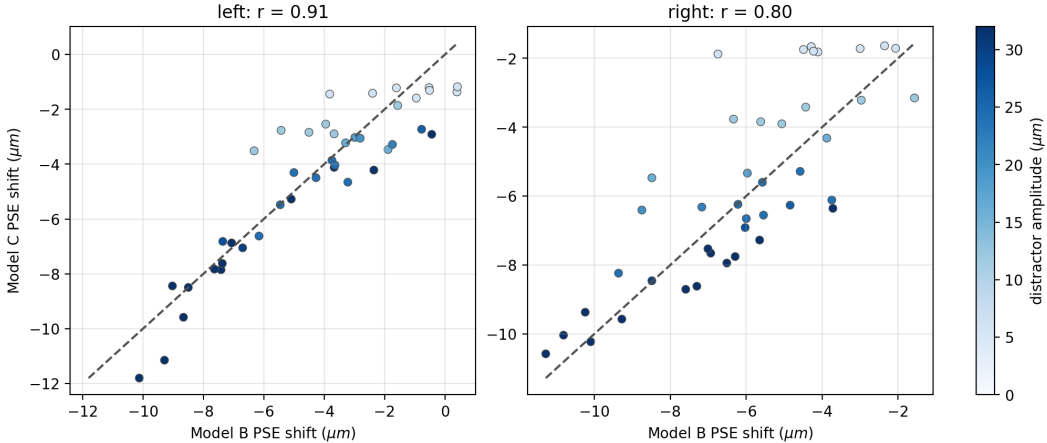
	selects A	selects B
data from A	9	1
data from B	0	10

- On data from A, B ties or loses: flexibility alone does not make it win
- The real data look like the data from B: sessions really differ

Model recovery: Δ in every replicate



Model C vs Model B: session PSE shifts



Model comparison: conclusions

- **Sessions differ:** B predicts far better than A, and model recovery shows this is not B's flexibility alone
- **The models agree on the PSE shift:** about 4 μm (left) and 6 μm (right) on average
- **B predicts better than C mainly because of lapses:** C's one lapse rate for both conditions misses the extra "high" responses to weak stimuli on bimanual trials
- Less clearly: a distractor-presence effect (right hand) and a bimanual slope

$\Rightarrow$ Model B is the model to use for inference for now

Comparison of the models

	A	B	C
Sessions	pooled	hierarchical	hierarchical
Distractor	separate groups	separate groups	covariate $\beta_2 x_{\text{dist}}$
Prediction (LOO)	3rd	**best**	2nd
PSE shift, left / right (μm)	-4.2 / -6.1	-4.3 / -6.1	-4.5 / -5.7

- All three models agree on the PSE shift
- B predicts best: sessions differ (B $\gg$ A), and lapse rates differ between conditions (B $>$ C)

Conclusions

- The distractor lowers the PSE: by about 4.2–4.5 μm (left) and 5.7–6.1 μm (right), in all three models.
- It raises the rate of "high" responses to weak stimuli on bimanual trials (γ_h); whether this grows with distractor amplitude is open.
- The sessions differ; a hierarchical model is needed.
- The shift grows with distractor amplitude, and a weak distractor may shift the PSE more than a proportional model allows (right hand).

Open questions

- **The extra "high" responses to weak stimuli on bimanual trials:** they seem to cluster in sessions with a strong distractor (being checked). What could produce them: responses to the distractor hand, attention, motor errors?
- **Amplitude and time:** the distractor amplitude rose over the sessions, so a growing effect could also be training or adaptation. Was the schedule chosen for a reason? Did anything else change over those months?
- **Left vs right:** the right hand shifts more, even at weak distractors. Is a hand asymmetry expected?

Next steps

- A new Model D, revising Model C: lapse rates per condition, testing whether the bimanual γ_h depends on distractor amplitude (or session order).
- A distractor-presence term only if it improves prediction.
- Model recovery for B against D.